LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 13 Test: Our Environment

Chapter / Topic: Our Environment

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SECTION A

Q1. Which group contains only biodegradable items? **[1]**

Answer: (b) Fruit peels, cake and lime juice

Q2. Which of the following forms a food chain? **[1]**

Answer: (c) Grass → goat → human

Q3. Which of the following are environment-friendly practices? **[1]**

Answer: (d) All of the above

Q4. In a food chain, the energy passed on to the next trophic level is about: **[1]**

Answer: (a) 10% of the previous level (10 per cent law)

Q5. Depletion of the ozone layer is mainly caused by: **[1]**

Answer: (b) chlorofluorocarbons (CFCs)

Q6. **Assertion (A):** Food chains generally have only three or four trophic levels. **Reason (R):** The energy available decreases greatly at each successive… **[1]**

Answer: (a). After a few levels, too little energy is left to support another level.

SECTION B

Q7. What are decomposers? What would happen if there were no decomposers? **[2]**

Answer: Decomposers are micro-organisms, such as bacteria and fungi, that break down the dead remains and waste products of organisms into simpler substances. Without them, dead bodies and wastes would pile up, and nutrients would not be returned to the soil for plants to use.

- Definition with examples: 1
- Consequence: 1

OR, answer: High-energy ultraviolet (UV) radiation splits oxygen molecules into free oxygen atoms: $O_2 \rightarrow O + O$. Each free oxygen atom combines with an oxygen molecule to form ozone: $O + O_2 \rightarrow O_3$.

- Each step: 1

Q8. What is biological magnification? Why is it highest in organisms at the top of a food chain? **[2]**

Answer: The progressive build-up of non-biodegradable chemicals, such as pesticides, at each trophic level of a food chain. Organisms at the top eat many organisms from the levels below, and these chemicals are not broken down, so the concentration keeps increasing up the chain.

- Definition: 1
- Reason: 1

SECTION C

Q9. Study the food chain: Grass → Grasshopper → Frog → Snake → Eagle **[3]**

(i) Producer: grass. Tertiary consumer: snake. [1]

(ii) 10 J (10,000 → 1,000 → 100 → 10 J, by the 10 per cent law) [1]

(iii) The eagle. It is at the top of the food chain, and pesticides build up at each level (biological magnification). [1]

Q10. Why is damage to the ozone layer a cause for concern? What steps have been taken to limit it? **[3]**

Answer: The ozone layer shields the Earth from harmful ultraviolet radiation from the Sun. UV radiation can cause skin cancer and harm other living things. Ozone levels fell sharply in the 1980s because of CFCs, which are used as refrigerants and in fire extinguishers. In 1987, UNEP succeeded in forging an agreement to freeze CFC production at 1986 levels, and companies now make CFC-free refrigerators.

- Shields from UV: 1
- Harmful effect / CFCs as cause: 1
- Steps taken: 1

OR, answer: (Any three) Separate biodegradable and non-biodegradable waste. Compost kitchen waste. Carry cloth bags instead of plastic bags. Avoid single-use plastic items. Reuse containers. Send paper, glass and metal for recycling.

- Any three: 1 each

SECTION D

Q11. Answer the following: **[5]**

(i) An ecosystem is made up of all the interacting organisms in an area together with the non-living parts of their environment. Biotic components are living things such as plants, animals and micro-organisms. Abiotic components are physical factors such as temperature, rainfall, wind, soil and minerals. [2]

- Definition: 1
- Biotic + abiotic with examples: 1

(ii) Green plants capture about 1% of the sunlight falling on their leaves and convert it into food energy. This energy passes from producers to herbivores to carnivores, but only about 10% is passed on to each next level. The rest is used for life processes or lost as heat. Energy that passes to the next level never comes back to the previous level, so the flow is in one direction only. [3]

- Plants capture sunlight: 1
- 10% passed on, rest lost: 1
- Never returns, so unidirectional: 1

OR, answer: (a) Biodegradable substances are broken down by bacteria and other decomposers (e.g. vegetable peels, paper, wood, cotton). Non-biodegradable substances are not broken down by biological processes (e.g. plastic, polythene, glass, DDT). (b) Any three: they stay in the environment for a long time; they pollute the soil and water; pesticides enter food chains and build up (biological magnification); plastic blocks drains and harms animals that swallow it.

- (a) difference 1, examples 1
- (b) any three: 1 each

SECTION E

Q12. Some years ago, it was proposed that tea on trains should be served in disposable kulhads (clay cups) instead of plastic cups. It was soon realised… **[4]**

(i) Plastic is non-biodegradable and pollutes the environment. [1]

(ii) Loss of fertile top soil. [1]

(iii) Over plastic: paper is biodegradable (and can be recycled). Over kulhads: making paper cups does not use up fertile top soil. [2]

- Each advantage: 1